PREDICTING FICTION AND NONFICTION CATEGORIES FROM BOOK DATA

Khang Nguyen, Huy Nguyen, Hung Nguyen

Introduction

This project predicts whether a book on Books to Scrape [1] belongs to the Fiction or Nonfiction category. The dataset has all 175 books from those two shelves, 65 Fiction and 110 Nonfiction, and the labels are the website's own categories. We did not include books from other shelves such as Mystery and Historical Fiction.

To find out which information helps predict the category, we compared titles and descriptions with cover colours, price and rating. The books already have labels, so this is a supervised learning task. We trained logistic regression on several feature sets and kept a separate test set to evaluate the model we selected.

Data scraping

We downloaded the HTML pages with Requests and found the elements we needed with Beautiful Soup 4 [2]. The scraper started from the website's category links and opened the Fiction and Nonfiction shelves. On each category page it collected the listed books, then followed the next-page link until it reached the last page of each shelf.

The category pages gave each book's full title, price, availability and rating, plus links to its product page and thumbnail. The label came from the shelf being scraped. The scraper then opened every product page and took the paragraph after the product-description heading. Relative links were resolved with urljoin.

Thumbnails were saved to an images folder. Everything else went into a CSV file, which stores each book's local image filename next to its title and category so the two stay paired. We kept the product-page link as a record ID: it let us check for duplicates and confirm that no book appeared in both the training and test sets, but it was not used as a predictor. We also collected availability, though every book was in stock, so it could not help the model.

Data cleaning and analysis

Our cleaning script (clean.py) checked for missing columns, blank values, prices that were zero, negative or not numeric, duplicate product links, and missing or empty image files. It can also check whether product links still load. The data passed every check: 175 records with no duplicate links, empty descriptions or missing required values, only positive prices, and 175 images that all opened and verified in Pillow with check_data.py [4]. We did not have to drop any records.

We split the data with a stratified 80/20 split (random state 42). The training set has 140 books, 52 Fiction and 88 Nonfiction, and the test set has 35, with 13 Fiction and 22 Nonfiction. Stratifying keeps the share of each category in both sets close to the full dataset.

All exploratory plots and comparisons used the training set only, so patterns in the test set could not affect which features we chose. Figure 1 shows category counts, price distributions and ratings. Prices overlapped between the categories, and both had books at every rating level from one to five.

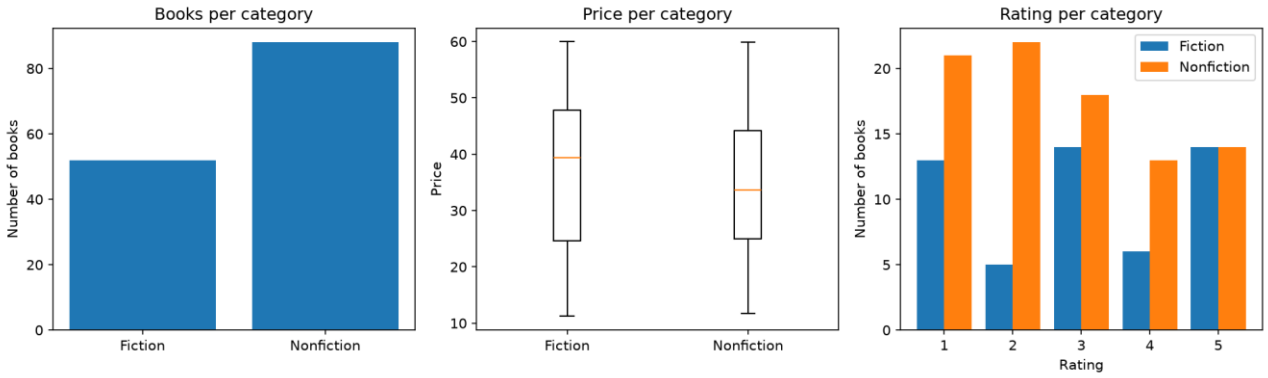


Figure 1. Exploratory analysis of the 140 training books. Prices in GBP.

The unequal category sizes also matter for evaluation. Predicting Nonfiction for every test book gives 62.9% accuracy while missing every Fiction book. We therefore used balanced accuracy as the main metric. It averages the recall of the two categories, so each category counts equally.

Feature engineering

For the text features we joined each title to its description. CountVectorizer lowercased this text, removed English stop words and counted individual words, giving a matrix with one column per vocabulary term. Bag-of-words ignores word order, so two books on similar topics can look alike to the model even when their categories differ.

We tested minimum document frequencies (min_df) of 1, 2 and 3. With min_df 3, a word is kept only if it appears in at least three of the books used to fit the vectorizer, which removes rare words and shrinks the feature set. In cross-validation the vocabulary was fitted inside each fold on that fold's training books, so validation books could not add terms or change which words passed the threshold.

For the image experiments we converted each cover to RGB with Pillow [4]. For each colour channel we calculated the mean and a four-bin histogram, and we added overall brightness, giving 16 features per cover. Histogram counts were divided by the number of pixels so covers of different sizes could be compared. These features describe how colour is distributed on a cover and say nothing about the objects or text printed on it.

The numeric features were price and rating, with rating words mapped to the integers 1 to 5. We standardized the numeric and image features using means and standard deviations from the training part of each fold.

We compared text alone, images alone and price and rating alone, then two combinations: text with images, and text with images and numeric features. The combined models used min_df 3 to limit the size of the text vocabulary. We also calculated title character counts, word counts and a digit indicator but did not include them in these comparisons.

Model training

We trained and evaluated the models with scikit-learn [3]. Logistic regression handles sparse word-count matrices and lets us inspect the learned word coefficients [5], and its probability estimates gave us ROC-AUC. We used the same classifier settings for every feature set: balanced class weights, because Nonfiction books outnumber Fiction books, a maximum of 1,000 iterations, and the default regularization strength C = 1.0.

For model selection we ran five-fold stratified cross-validation on the 140 training books. The folds were shuffled with random state 42 and reused for every feature set. Each fold fitted on 112 books and validated on 28, and the text vocabulary and any scaling parameters were fitted on those 112 books alone. A majority-class DummyClassifier was the baseline.

Table 1 gives balanced accuracy, with its standard deviation across the five folds, and macro-F1. Both metrics weight the two categories equally.

Features	Mean vocabulary size	Balanced accuracy mean (SD)	Macro-F1 mean
Majority baseline	0	50.0 (0.0)	0.386
Price and rating	0	52.7 (4.7)	0.517
Image features	0	62.6 (11.3)	0.616
Text with min_df 1	5,326	73.2 (7.3)	0.739
Text with min_df 2	1,714	75.3 (6.4)	0.761
Text with min_df 3	879	76.9 (9.4)	0.769
Text with min_df 3 and images	879	77.1 (10.7)	0.770
Text with min_df 3, images and numeric features	879	76.1 (9.5)	0.762

Table 1. Training-set cross-validation. Balanced accuracy is shown as a percentage, with the sample standard deviation across folds.

Text with images had the highest mean balanced accuracy, 0.2 percentage points above text alone with min_df 3, a much smaller gap than the variation between folds. We did not want to choose the larger feature set on that basis, so we used a rule of thumb: among feature sets whose mean was within one standard deviation of the best mean, pick the simplest. The standard deviation came from the best-scoring set, and we did not run a significance test. This rule selected text alone with min_df 3.

Compared with min_df 1, the selected text features used about 83.5% fewer vocabulary terms and had a higher observed mean score, which suited a dataset this small. Five folds are not enough to tell how large that difference really is. Price and rating scored close to the baseline. Image features did better on their own but added little when combined with text.

Model evaluation

After selection, we fitted the chosen vectorizer and logistic regression on all 140 training books. The vocabulary grew to 1,112 terms, above the cross-validation average, because with more books more words reach the three-document threshold. We then evaluated the model on the 35 test books, encoding Nonfiction as label 1 and computing ROC-AUC from its predicted probabilities.

Table 2. Results on the separate test set. The baseline always predicts Nonfiction and assigns it a constant score.

Metric	Selected model	Majority baseline
Accuracy	71.4%	62.9%
Balanced accuracy	66.3%	50.0%
Macro-F1	0.669	0.386
ROC-AUC	0.857	0.500

The model classified 25 of the 35 test books correctly, against 22 for the baseline. As Figure 2 shows, it found 6 of the 13 Fiction books (46.2% recall) and 19 of the 22 Nonfiction books (86.4% recall). Fiction recall stayed low even with balanced class weights.

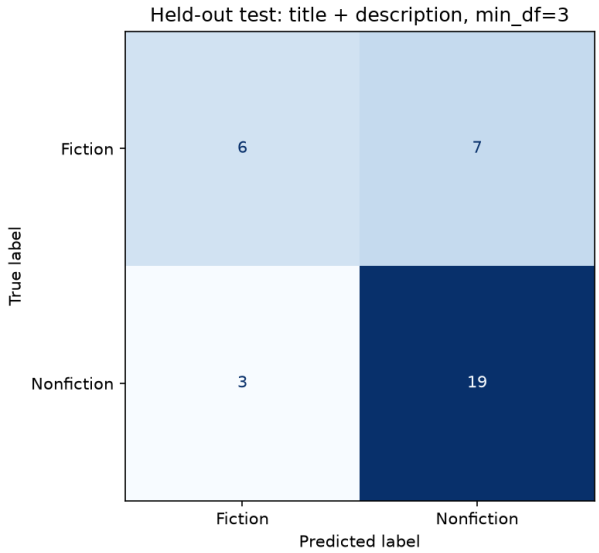


Figure 2. Test-set confusion matrix.

Test balanced accuracy was lower than the cross-validation mean of 76.9%. The test set is small, so each book moves the scores a lot: one more correct prediction changes accuracy by 2.9 percentage points, and one more correct Fiction book changes Fiction recall by 7.7. The cross-validation standard deviation measures variation between folds and is not a confidence interval for the test score.

ROC-AUC was 0.857. It measures how well the predicted probabilities rank Nonfiction books above Fiction books, while balanced accuracy depends on the decisions made at the default 0.5 threshold. Four of the seven missed Fiction books had Nonfiction probabilities between 0.51 and 0.66, just over the threshold. We kept the threshold at 0.5, since setting it from the test labels would bias the test result.

We read the descriptions of the ten misclassified books to look for explanations. Most Fiction books predicted as Nonfiction described something from the real world, such as real places, historical events or money (I Am Pilgrim, Cometh the Hour, Mothering Sunday, The Firm and Memoirs of a Geisha). The Nonfiction books predicted as Fiction were either told as a story (Into the Wild, A Murder Over a Girl) or used relationship vocabulary (The Five Love Languages). “Novel” had the strongest coefficient towards Fiction, yet Memoirs of a Geisha was misclassified even though its description contains the word, because the other words in it outweighed that cue. The ten strongest words for each category also included “don” (from “don’t”) and “la”, which are tokenization artifacts. Ten examples cannot pin down the cause of each error, but they suggest the model learned topic vocabulary more than narrative form, so Fiction set in the real world and Nonfiction written as a story are the hardest cases.

Conclusion and discussion

Titles and descriptions were the most useful features in our experiments. Logistic regression with min_df 3 reached 71.4% accuracy and 66.3% balanced accuracy on the test set, above the majority baseline. It still misclassified ten of the 35 test books, most of them Fiction books predicted as Nonfiction.

Removing rare words reduced the number of features and gave a higher observed cross-validation mean. Cover-colour features added little to the text model. They ignore the layout and content of the cover, so a richer image representation is one thing future work could test.

The dataset is small and comes from two shelves of one demonstration website, and its labels follow that site's organisation, so the results may not hold for other books. More data, especially more Fiction books, would give more reliable performance estimates. We compared features with a single classifier at a fixed regularisation strength, which does not show whether logistic regression is the best classifier for this task. Future experiments could try Multinomial Naive Bayes, other values of C or TF-IDF weighting, tuned on the training data and evaluated on a new, independent test set.

References

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AI Use: We used ChatGPT and Claude to help debug, edit, polish and format the code and the report.